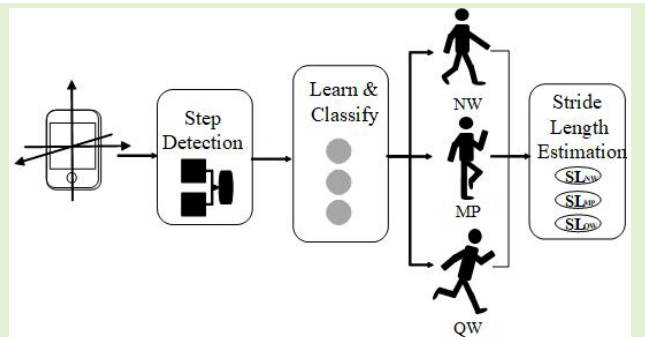


A Robust Step Detection and Stride Length Estimation for Pedestrian Dead Reckoning Using a Smartphone

Yingbiao Yao¹, Lei Pan, Wei Feng¹, Xiaorong Xu¹, *Member, IEEE*, Xuesong Liang, and Xin Xu

Abstract—As an infrastructure-free positioning and navigation method, pedestrian dead reckoning (PDR) is still a research hotspot in the field of indoor localization. Step detection (SD) and stride length estimation (SLE) are two key components of PDR, and it is a challenging problem to apply SD and SLE to different walking patterns. Focusing on this problem, this paper proposes a robust SD and SLE method based on recognizing three walking patterns (i.e., Normal Walk, March in Place, and Quick Walk) using a smartphone. First, we propose a dynamic time warping–based peak prediction with zero-crossing detection to improve the SD accuracy. In particular, the proposed SD can accurately identify the starting and ending points of each step in the three walking patterns. Second, according to the extracted features of each step, a random forest algorithm with classification proofreading is used to recognize the three walking patterns. Finally, an improved SLE model is proposed for the different walking patterns to achieve a higher SLE accuracy. The experimental results show that, on average, the SD accuracy is about 97.9%, the recognition accuracy is about 98.4%, and the relative error of the estimated walking distance is about 3.0%, which outperforms those of the existing commonly used SD and SLE methods.

Index Terms—Pedestrian dead reckoning, step detection, stride length estimation, walking pattern.



I. INTRODUCTION

ALTHOUGH global positioning system (GPS)–based [1] outdoor localization is very common, indoor localization remains a challenge because GPS signals are unavailable indoors. Many indoor localization methods are based on wireless radio systems, such as Wi-Fi [2], radio-frequency identity [3], ultra-wideband [4], and Bluetooth [5]. These localization methods can be categorized into two types [6]: triangulation and fingerprinting. The former relies on installed expensive hardware, making it neither scalable nor universal. The latter requires pretraining, which is time-consuming.

In addition to wireless radio-based methods, pedestrian dead reckoning (PDR) [7], [8], relying on inertial sensors, provides another way for indoor localization. PDR-based methods are advantageous, as they derive the current location by adding the estimated displacement to the previous location, independent

of the external infrastructure. PDR localization errors mainly depend on the accuracy of the step detection (SD), stride length estimation (SLE), and heading direction estimation.

Nowadays, it is still a challenging problem to adapt SD and SLE to different walking patterns. For example, when the pedestrian is in the state of Marching in Place (MP), the traditional SD method has difficulty counting the steps because there are multiple close peaks or valleys in each step. Moreover, when the pedestrian is in the state of MP, Normal Walking (NW), or Quick Walking (QW), the SLE model should be different because these modes have different walking characteristics (e.g., the stride length of the MP pattern should be zero).

The purpose of this paper is to design a robust SD and SLE method that can adapt to the changes in walking patterns, phone poses, and users. Specifically, we consider two phone poses in which the phone is held in front of the chest (Pose1) or put in a trouser pocket (Pose2) and three walking patterns: NW, MP, and QW. NW is the most common walking pattern, and its speed range is usually 1–1.5 m/s. In MP, the pedestrian's two feet move up and down while remaining in place, and it does not cause walking distance. In QW, the pedestrian walks or jogs at a brisk pace, and its speed range is usually 2–2.5 m/s.

Based on the above considerations, we propose a robust SD and SLE method for PDR using a smartphone's built-in

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inertial sensors. The key idea behind the proposed method is to recognize the pedestrian's walking patterns and then to select the SLE model according to the recognized walking patterns, which can improve the existing SD and SLE methods' robustness and practicability. To our knowledge, the contributions of this paper can be summarized as follows:

- We propose a robust SD method, which combines a dynamic time warping (DTW)-based peak prediction with zero-crossing detection, to accurately detect the step time boundaries in different walking patterns.
- We propose a random forest (RF)-based walking pattern-aware SLE method to improve the SLE accuracy which, using new features and classification proofreading (CP), can recognize the walking patterns of each step, and then use our improved SLE models for different walking patterns.
- We compare our method with the existing SD and SLE methods and demonstrate that our method outperforms the commonly used methods.

The rest of the paper is organized as follows. Section II introduces the related work. The components and design details of our proposed method are described in Section III. Then, we present the experiment results and analysis in Section IV. Finally, Section V concludes this paper.

II. RELATED WORK

A. Step Detection

Generally, the existing SD algorithms fall into 3 categories: time-domain methods [9]–[14], frequency-domain methods [15]–[17], and feature-clustering methods [18]–[20]. Since our method belongs to the time-domain methods, we focus on the SD methods in the time domain. The time-domain methods include threshold detection [9], peak detection [10], zero-crossing detection [11], correlation detection [12], etc. Threshold detection counts steps by judging whether sensor data surpasses predefined thresholds. However, though the thresholding method is the simplest, it is difficult to preset an optimal threshold for all cases. Peak detection estimates steps based on the number of peaks in a sequence of sensor data, but it suffers from interference peaks due to noise and occasional disturbance. Zero-crossing detection measures steps by detecting the number of zero points in sensor data, but there are many false zero points due to interference and noise. Nowadays, due to the complexity of actual sensor signals, it is difficult to achieve satisfactory SD accuracy with only one method. Therefore, combination methods are proposed to improve the SD accuracy. In [13], Zhang *et al.* combined zero-crossing and peak detection to limit the interval between two zero points and reduce false zero points. In [14], the DTW method [21] is adopted for evaluating the correlation of three-axis accelerations, and it can limit the problems resulting from naturally occurring changes in walking speed.

B. Walking Pattern Recognition

Using inertial sensors to sense human activity, including pedestrian walking patterns, has been explored in academia.

Susi *et al.* [22] proposed a decision tree-based classifier to recognize four motion modes. WTrack proposed a hidden Markov model-based walking pattern recognition [23] with a total of 72 patterns, in which walking patterns are the combination of walking speed and direction. In [24], with a support vector machine and decision tree, combinations of motion states and phone poses can be recognized successfully. Catal *et al.* [25] utilized the J48 decision tree, multilayer perceptron, and logistic regression techniques and combined these classifiers with the average of probabilities combination rule to recognize human activities. Miyamoto and Ogawa [26] used histograms of oriented gradients to extract features and used Real AdaBoost as a classifier. In [27], the authors demonstrated the feasibility of using the newly emerging barometer in modern smartphones to distinguish vertical walking patterns, such as upstairs and downstairs. Recently, deep learning-based activity recognition has been getting a lot of attention. Hassan *et al.* [28] proposed a deep belief network-based approach for human activity recognition. Gu *et al.* [29] adopted a stacked denoising autoencoder-based deep learning method for locomotion activity recognition using data from multiple sensors available on most smart devices. In [30], the authors employed a deep learning method to recognize different walking speeds and phone-carrying methods using stacked autoencoders. Zhou *et al.* [31] proposed smartphone-based activity recognition using a new convolutional neural network, which can identify nine activities for use as landmarks for indoor localization.

C. Stride Length Estimation

Assuming stride length is proportional to the vertical displacement of the hip at each step, Weinberg proposed a famous SLE model in [32], which is based on the maximum and minimum acceleration within each step. Alvarez *et al.* [33] modeled stride length as a function of height and leg length by considering human gait as an inverted pendulum model. Wei *et al.* [34] proposed an SLE system based on computer vision to assess the movement of Parkinson's patients. In [35], an SLE method based on the wavelet transform is presented. Shin *et al.* [36] proposed an adaptive SLE method according to the pedestrian's motion status, that is, walking or running. Double integration is another important approach for SLE algorithms [37]. However, due to intrinsic measurement errors in the currently available inertial sensors, minimizing the integration drift of low-precision sensors in smartphones remains an open question. With the recent advances in machine learning, several machine learning-based SLE approaches have been proposed. For example, a deep convolutional neural network and an online sequential extreme learning machine were used to estimate stride length in [38], [39] and [13], respectively.

III. THE PROPOSED METHOD

The architecture of our method, shown in Figure 1, consists of four modules: data preprocessing, SD, walking pattern recognition (WPR), and SLE. The contributions of this paper cover three modules: SD, WPR, and SLE. In the SD module,

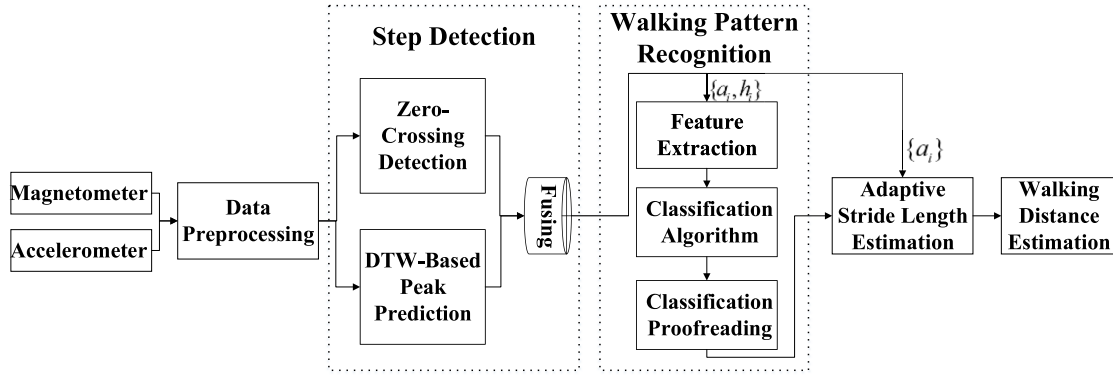


Fig. 1. The architecture of our method.

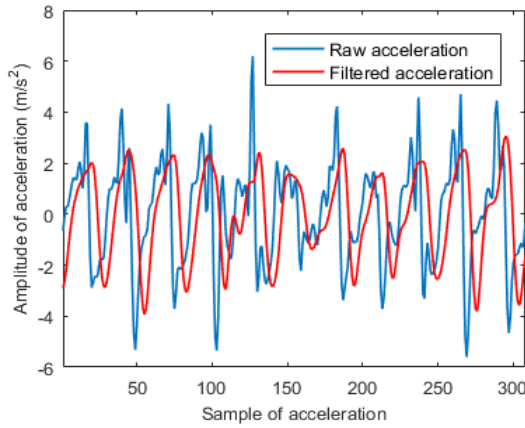


Fig. 2. The performance of the 4th-order Butterworth filter.

a DTW-based peak prediction is combined with zero-crossing detection to accurately detect the step boundaries. In the WPR module, the RF algorithm with the new features and CP is used to recognize the walking patterns. In the SLE module, two state-of-the-art SLE methods are fused.

A. Data Preprocessing

Data preprocessing is used to ease noise and interference effects. The original derived accelerometer data includes X, Y, and Z values, and we use the acceleration amplitude to detect the step event, as follows:

$$a_t = \sqrt{a_{x,t}^2 + a_{y,t}^2 + a_{z,t}^2} - g \quad (1)$$

where $a_{x,t}$, $a_{y,t}$, and $a_{z,t}$ are the accelerometer readings of the X-axis, Y-axis, and Z-axis acquired at time t , respectively, and g is the gravity component, equal to 9.8 m/s^2 .

The raw acceleration may contain interference and noise signals, which can deteriorate the subsequent SD and WPR performance. Therefore, a 4th-order Butterworth digital low-pass filter, with a cutoff frequency of $w_c = 0.2\pi$, is applied to filter the high frequency component of a_t , and its filtering effect is shown in Figure 2. It should be pointed out that although there is a delay in the filtered signal in Figure 2, this delay has no other effect except that the output results lag.

B. Step Detection

The goals of the SD include: 1) it can meet the requirements of step counting under different walking patterns, and 2) it can correctly segment the data of each step, laying a foundation for the subsequent walking pattern recognition. To achieve these two goals, the proposed SD method combines DTW-based peak prediction with cross-zero detection. The former realizes accurate step counting under different walking patterns, and the latter can identify the starting point of each step.

1) **Zero-Crossing Detection:** Park *et al.* [40] showed that arm and foot movements are synchronized during walking. Therefore, the starting or ending point of each step should be a zero point in the acceleration signal, according to the gait model in [41]. However, due to the influence of the sampling rate, walking speed, and noise, the sampled acceleration signal may have no zero point, or the zero point may not be the true starting or ending point. Hence, we recorded the point passing through the zero as the zero point. All potential starting and ending points, determined by zero-crossing detection, are marked as set $S1$, as shown in (2):

$$S1 = \{t | a_t \times a_{t-1} \leq 0\} \quad (2)$$

However, the zero-crossing detection method generally has a problem with false zero points. Therefore, we combine zero-crossing detection with peak detection to exclude false zero points from the set $S1$.

2) **DTW-Based Peak Prediction:** Peak detection is a common method for SD. However, it also suffers from false peaks, especially when a pedestrian is in the MP pattern. An example is shown in Figure 3, in which the number of steps using the traditional peak detection method will be much greater than the real number, due to false peaks. To solve this problem, the DTW-based peak prediction method is proposed to calculate the similarity of two adjacent steps' acceleration signals and then predict the range of the true peak.

The DTW-based peak prediction method consists of three key modules: the periodicity constraint, DTW constraint, and searching maximum peak, as shown in Figure 4.

a) **Periodicity constraint:** In general, the periods of two adjacent steps in the same walking pattern have similarities. Here, the period refers to the difference between two

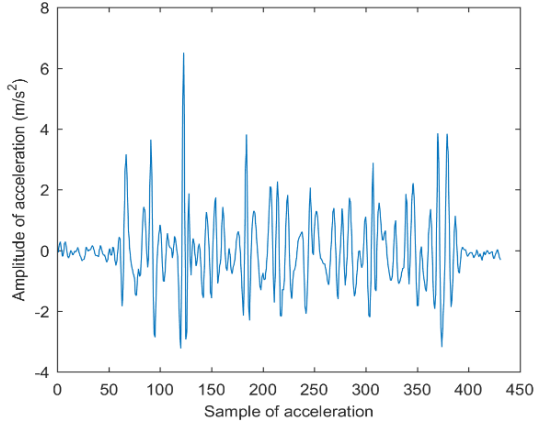


Fig. 3. Acceleration waveform of 10 steps in the MP pattern.

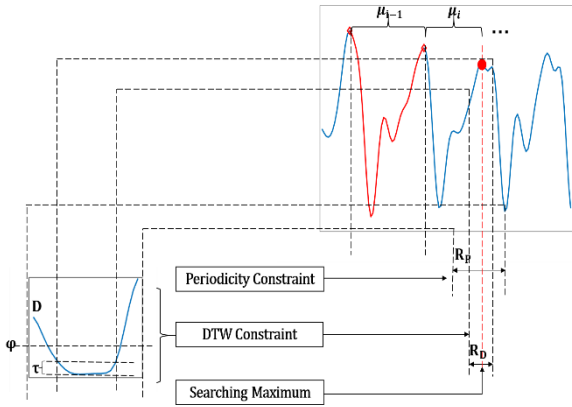


Fig. 4. The DTW-based peak prediction.

consecutive true peak moments, which is defined as follows:

$$t_i = t_{i+1}^{peak} - t_i^{peak} \quad (3)$$

where t_i^{peak} is the moment of the i -th peak, and t_i is the period of the i -th step.

The periods' similarity indicates that each period in the same walking pattern is very close to the next. Therefore, the next step's period should satisfy the following periodic similarity constraint:

$$t_i \in ((1 - \theta) * T_{i-1}, (1 + \theta) * T_{i-1}) \quad (4)$$

where θ is the similarity parameter, and T_i is the period template. To improve the accuracy of T_i , it is updated as follows:

$$T_i = \frac{T_{i-1} + t_i}{2} \quad (5)$$

According to the periodic constraints, we can initially derive the range of the next peak, $\mathbf{R}_p$, as shown in Figure 4.

b) *DTW constraint*: DTW was proposed to calculate the distance or similarity of two sequences with different lengths in the time domain [21]. In this paper, we employed DTW to predict the time range of the current step's true peak acceleration.

Algorithm 1 DTW-Based Peak Prediction Method

Input: The acceleration $\{a\}$, the latest two peak moments t_{i-1}^{peak} , t_i^{peak} , the period template T_{i-1} .

Output: The next peak moment t_{i+1}^{peak} , the next period template T_i .

1. $\mathbf{R}_p = \emptyset$; /* Initialize $\mathbf{R}_p$ */
2. **FOR** each moment j
3. **IF** $j - t_i^{peak} \in ((1 - \theta) * T_{i-1}, (1 + \theta) * T_{i-1})$
4. $\mathbf{R}_p = \mathbf{R}_p \cup \{j\}$
5. **ENDIF**
6. **ENDFOR**
7. $\mu_{i-1} = a(t_{i-1}^{peak} : t_i^{peak})$; /* Extract the matched sequence */
8. **FOR** each moment k of $\mathbf{R}_p$
9. $v_k = a(t_i^{peak} : k)$; /* Build the matching sequence */
10. $\mathbf{D}(k) = DTW(\mu_{i-1}, v_k)$
11. **ENDFOR**
12. $\mathbf{R}_D = \emptyset$; /* Initialize $\mathbf{R}_D$ */
13. **FOR** each moment k of $\mathbf{R}_p$
14. **IF** $\mathbf{D}(k) < \varphi$ & $\mathbf{D}(k) - \min(\mathbf{D}) < \tau$
15. $\mathbf{R}_D = \mathbf{R}_D \cup \{k\}$
16. **ENDIF**
17. **ENDFOR**
18. $t^{peak} = \arg \max_{t \in \mathbf{R}_D} (a(t))$
19. $t_i = t_{i+1}^{peak} - t_i^{peak}$
20. $T_i = \frac{T_{i-1} + t_i}{2}$

We consider the acceleration sequence between the latest two peaks as the matched sequence μ , as shown in (6). In particular, the first matched sequence μ_0 is obtained using the traditional peak detection method.

$$\mu_{i-1} = a(t_{i-1}^{peak} : t_i^{peak}) \quad (6)$$

We take the acceleration data from the latest peak to the first sample point in $\mathbf{R}_p$ as the first test sequence v_1 , the acceleration data from the latest peak to the second sample point in $\mathbf{R}_p$ as the second test sequence v_2 , and so on, to obtain a test sequence set $\mathbf{V} = \{v_1, v_2, \dots, v_j, \dots, v_k\}$. Then, the DTW distance of μ and v_j , $DTW(\mu, v_j)$, is calculated to obtain a DTW distance set $\mathbf{D} = \{DTW(\mu, v_1), DTW(\mu, v_2), \dots, DTW(\mu, v_k)\}$. According to the similarity of each step in the same walking pattern, the DTW distance will be significantly small when the latest sample point in the test sequence is near the next true peak. Hence, the sample point j in $\mathbf{R}_p$, which satisfies the condition in (7), is outputted to the candidate true peak set, $\mathbf{R}_D$, which is the prediction range of the next true peak.

$$\mathbf{R}_D = \{j | j \in \mathbf{R}_p, DTW(\mu, v_j) < \varphi,$$

$$DTW(\mu, v_j) - D_{\min} < \tau\} \text{ and } D_{\min} = \min(\mathbf{D}) \quad (7)$$

where φ and τ are used to control the number of elements in $\mathbf{R}_D$. In addition, since the DTW constraint is to utilize the

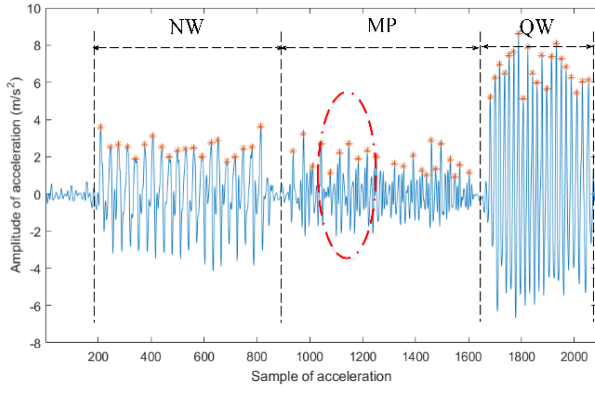


Fig. 5. Results of the DTW-based peak prediction.

TABLE I
PROCESS OF DTW-BASED PEAK PREDICTION

Moment of current peak	Prediction range of next peak
1043	(1071,1083)
1077	(1106,1120)
1113	(1143,1152)
1147	(1170,1186)
1183	(1207,1223)
1217	(1244,1260)

similarity of each step in the same walking pattern, the $\mathbf{R}_D$ will be empty when the pedestrian changes the walking pattern.

c) *Searching Maximum Peak*: After obtaining the next true peak's prediction range, $\mathbf{R}_D$, the maximum acceleration in $\mathbf{R}_D$ corresponds to the true peak moment, as shown in (8).

$$t^{peak} = \arg \max_{t \in \mathbf{R}_D} (a(t)) \quad (8)$$

where t^{peak} represents the current true peak moment.

Algorithm 1 shows the process of the DTW-based peak prediction in detail, and its improvements are: 1) the parameters θ , φ , and τ are the same for the NW, MP, and QW patterns; 2) T_i and μ are automatically established and updated according to the similarity of each step in the same walking pattern; 3) once a pattern change is detected, μ_0 will be reestablished. Therefore, our DTW-based peak prediction method can adapt to different walking patterns.

An example is given in Figure 5 for detecting the true peak in three walking patterns. From Figure 5, we find that the DTW-based peak prediction method accurately detects the peak in NW, MP, and QW.

Furthermore, Table I shows the detailed peak prediction at the red circle marked in Figure 5. When the current peak moment is 1043, we obtain the prediction range of the next peak (1071, 1083), from which 1077 is searched as the moment of the next peak. Then we take 1077 as the current peak moment to predict the range of the next peak, and so on. This process will be repeated until all peaks are accurately detected.

3) *Fusing*: In this paper, the peak moment is obtained using the DTW-based peak prediction method, while the valley moment is the moment corresponding to the minimum acceleration value between two peak moments. Then, the peak

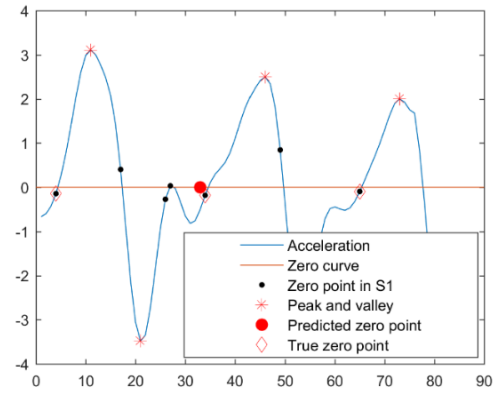


Fig. 6. Example of eliminating the false zeros.

and valley moments are used to predict the true zero-point moment, t_{zero}^i , which is defined in (9):

$$t_{zero}^i = \frac{(t_{i-1}^{valley} + t_i^{peak})}{2} \quad (9)$$

where t_i^{peak} and t_{i-1}^{valley} are the current peak and last valley moments, respectively.

As depicted in Figure 6, we obtain the true zero-point set, $\mathbf{S}_2$, through finding the point in $\mathbf{S}_1$ that is the closest to the predicted zero point, where $\mathbf{S}_1$ is the set of zero points determined by zero-crossing detection, as shown in Equ. (2). Assuming that $\mathbf{S}_1 = \{t(1), t(2), t(3), \dots, t(i), \dots\}$ and t_{zero} is a predicted zero point, then t is a true zero point:

$$\mathbf{S}_2 = \left\{ t \mid t = \arg \min_{t(i) \in \mathbf{S}_1} (|t(i) - t_{zero}|) \right\} \quad (10)$$

Then the starting and ending points of each step can be expressed as $\{t_i, t_{i+1}\}$, where $t_i, t_{i+1} \in \mathbf{S}_2$, $i = 1, 2, 3, \dots$. From each pair of starting and ending points, we can extract each step's complete information, which forms the basis of feature analysis and feature extraction for WPR.

C. Walking Pattern Recognition

1) *Feature Extraction*: Based on the step detection above, the data of each step from the starting point to the ending point can be obtained. Then, the following features, shown in Table II, are extracted. Moreover, apart from the acceleration data, each step's magnetometer data is also used for feature extraction to increase the recognition accuracy. In Table II, a and h are the amplitude of the acceleration and magnetometer data, respectively; p is the amplitude of the acceleration data power spectrum; $\mathbf{E}[\cdot]$ is for expectation; $\mathbf{I}[\cdot]$ indicates function; and $\mathbf{I}[\cdot] = 1$ when the condition is true; otherwise it is 0.

Some features in Table II are from [28], [42]–[45], and some features (i.e., F1, F8, F9, F12–F15, and F22–F29) are newly proposed in this paper. Here, we only introduce four newly proposed features that pass the subsequent feature selection: F1 (μ_a), F13 (FC), F12 ($diff_{median}$), and F8 (ρ_{zero}). These four features are all extracted from the acceleration signal. Specifically, μ_a is the mean of the absolute value, FC is the center frequency of the power spectrum amplitude, $diff_{median}$ is the difference between the peak and median, and

TABLE II
THE EXTRACTED FEATURES

Feature index	Definition
F1	The mean of the absolute acceleration value, $\mu_a = \text{mean}(a_i)$
F2	The acceleration variance, $\sigma_a^2 = \text{var}(a_i)$ [28]
F3	The acceleration skewness, $\text{ske}_a = E\left[\left((a_i - \text{mean}(a_i))/\sqrt{\text{var}(a_i)}\right)^3\right]$ [28]
F4	The acceleration kurtosis, $\text{kur}_a = E\left[\left((a_i - \text{mean}(a_i))/\sqrt{\text{var}(a_i)}\right)^4\right]$ [28]
F5	The magnetic field strength variance, $\sigma_h^2 = \text{var}(h_i)$ [41]
F6	The magnetic field strength skewness, $\text{ske}_h = E\left[\left((h_i - \text{mean}(h_i))/\sqrt{\text{var}(h_i)}\right)^3\right]$
F7	The magnetic field strength kurtosis, $\text{kur}_h = E\left[\left((h_i - \text{mean}(h_i))/\sqrt{\text{var}(h_i)}\right)^4\right]$
F8	The number of zero points in each step, $\rho_{\text{zero}} = \sum_{j=1}^L \mathbb{I}[S1(j) \in (t_i, t_{i+1})]$
F9	The number of local maximum points in each step, $\rho_{\text{extreme}} = \sum_{t=t_i}^{t_{i+1}} \mathbb{I}[a_t > a_{t+1} \ \&\& \ a_t > a_{t-1}]$
F10	The difference between two successive peaks of acceleration, $\text{diff}_{\text{peak}} = a(t_{i+1}^{\text{peak}}) - a(t_i^{\text{peak}})$
F11	The period duration, $\text{diff}_{\text{time}} = t_{i+1}^{\text{peak}} - t_i^{\text{peak}}$ [43]
F12	The difference of the peak and median in each step, $\text{diff}_{\text{median}} = a(t_i^{\text{peak}}) - \text{median}(a_i)$
F13	The center frequency, $FC = \sum_{i=1}^N f_i p_i / \sum_{i=1}^N p_i$
F14	The mean square frequency, $MSF = \sum_{i=1}^N f_i^2 p_i / \sum_{i=1}^N p_i$
F15	The frequency variance, $VF = \sum_{i=1}^N (f_i - FC)^2 p_i / \sum_{i=1}^N p_i$
F16-F18	The mean of the absolute value of acceleration of the X-axis, Y-axis, and Z-axis [40]
F19-F21	The variance of acceleration of the X-axis, Y-axis, and Z-axis [40]
F22-F24	The skewness of acceleration of the X-axis, Y-axis, and Z-axis
F25-F27	The kurtosis of acceleration of the X-axis, Y-axis, and Z-axis
F28	Correlation coefficient between X-axis and Y-axis acceleration,
F29	Correlation coefficient between X-axis and Z-axis acceleration,
F30	Correlation coefficient between Y-axis and Z-axis acceleration [42]

ρ_{zero} is the number of zero points in each step. F1 and F13 are effective at recognizing the smartphone pose and walking patterns. F12 and F8 are effective at recognizing the walking patterns.

2) Classification Algorithm: Aiming at the classification problem in this paper, we have implemented three classification algorithms: back propagation neural networks, support vector machine and RF. Experiment results show that RF has the advantages of high accuracy and low complexity, so RF is used to recognize the smartphone poses and walking patterns. We trained three RF classifiers in this paper, as shown in Figure 7. Classifier 1 is used to recognize the smartphone pose

(i.e., Pose1: holding in front of the chest, and Pose2: carrying in the trouser pocket, as shown in Figure 8). Classifiers 2 and 3 are used for WPR (i.e., NW, MP, and QW) in Pose1 and Pose2, respectively.

3) Feature Selection: To make a tradeoff between the accuracy and efficiency of the above three classifiers, we use the Relief-F method in [46], [47] to select the suitable features instead of adopting all 30 features in Table II.

The Relief-F results of the three classifiers are shown in Figure 9. We finally selected the following features, shown in Table III, for each classifier.

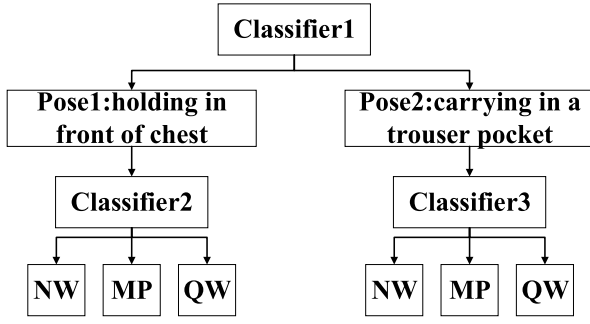


Fig. 7. The structure diagram of classifier 1.

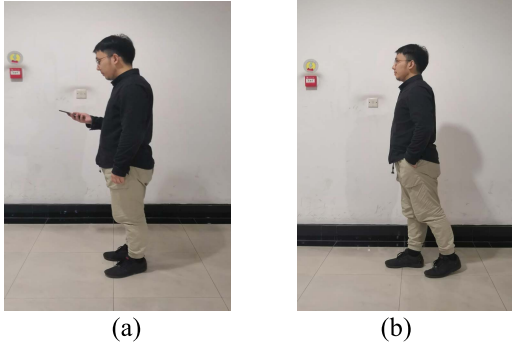


Fig. 8. Two phone poses: (a) holding in front of the chest and (b) carrying in a trouser pocket.

TABLE III
THE SELECTED FEATURE FOR EACH CLASSIFIER

Classifier	Selected features
Classifier1	F17, F18, F16, F13, F1, F20, F10, F14, F21, F23
Classifier2	F1, F13, F16, F19, F12, F8, F14, F2, F9, F11
Classifier3	F1, F20, F11, F2, F4, F12, F28, F29, F30, F16

4) Classification Proofreading: In general, a pedestrian's walking pattern lasts for a period of time instead of changing immediately. Therefore, the adjacent steps' recognition results can be used to improve the walking patterns' recognition accuracy. Specifically, if the recognition result of the current step i is different from that of step $i-1$, the recognition decision is given at most three steps later. We consider the recognition result of the subsequent steps $i+1$, $i+2$, and $i+3$. If they are the same as that of step i , then the recognition result of step i is credible, and we take the classifier result as the final walking pattern. Otherwise, we maintain the walking pattern of step $i-1$ in step i , as shown in (11):

$$WP(i) = \begin{cases} c(i) & \text{if } c(i) = c(i+1) \\ & = c(i+2) = c(i+3) \\ WP(i-1) & \text{otherwise} \end{cases} \quad (11)$$

where $WP(i)$ is the final recognized walking pattern of step i , and $c(i)$ is the classification result of step i . Equ. (11) illustrates that only when four consecutive steps are identified

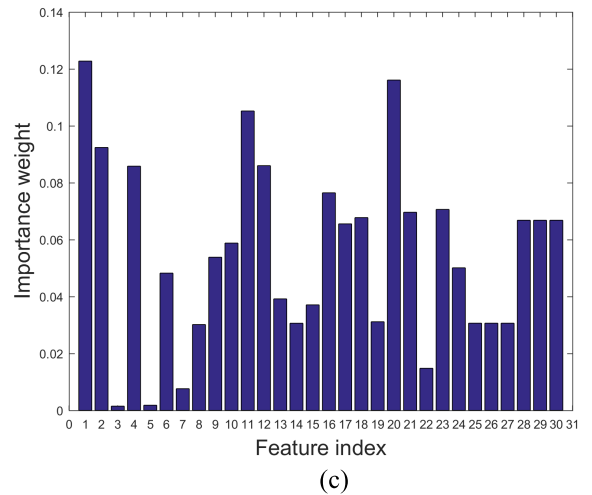
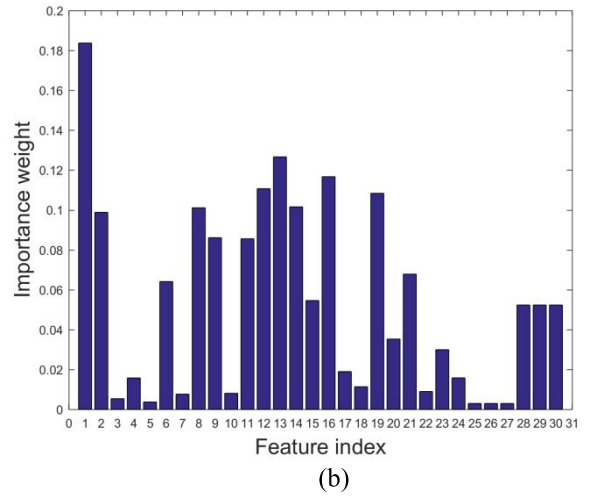
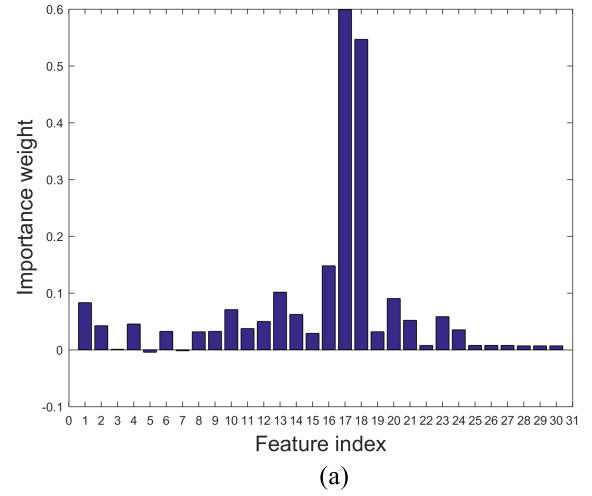


Fig. 9. The Relief-F results of the three classifiers: (a) classifier1, (b) classifier2, and (c) classifier3.

as the same walking pattern does our method consider the walking pattern to have changed.

D. Adaptive Stride Length Estimation

In general, stride length differs between persons and is related to a variety of factors, such as the pedestrian's height, weight, and physical condition. Even for the same person,

the stride length of each step in the walking process will be different. In particular, when a pedestrian changes his or her walking pattern, the change in stride length is also obvious. For example, the stride length should be zero when the pedestrian is in the MP pattern. Therefore, we propose a walking pattern-aware SLE method to adapt to three walking patterns: NW, MP, and QW. Moreover, to make the proposed SLE method irrelevant to specific pedestrians, the pedestrians' physiological parameters are not included in our proposed SLE method. Our proposed SLE model combines two popular SLE methods [36], [39], as shown in (12):

$$L = \alpha * \sqrt[4]{a_{\max,k} - a_{\min,k}} + \beta * f_k + \gamma * v_k \quad (12)$$

where α , β , and γ are step parameters that can be determined by the training data, $a_{\max,k}$ and $a_{\min,k}$ represent the maximum and minimum of the acceleration at step k , and f_k and v_k are the pedestrian's walking frequency and the variance of acceleration signals, respectively, which can be calculated as follows:

$$f_k = \frac{1}{t_{k+1} - t_k} \quad (13)$$

$$v_k = \frac{1}{N_k} \sum_{t=t_k}^{t_{k+1}} (a_t - \bar{a}_k)^2 \quad (14)$$

where t_k, t_{k+1} is the starting and ending time of step k , a_t is the acceleration at time t , $\bar{a}_k$ represents the average acceleration of step k , and N_k denotes the number of samples during step k .

The step parameters vary with the walking patterns, and each walking pattern corresponds to a set of step parameters. There is no doubt that the stride length of MP is 0 and, hence, α , β , and γ are all zero in the MP pattern. The step parameters in the NW and QW patterns can be obtained using the least mean square algorithm. When taking SL as the stride length in the real case, then the sum of squared errors of n steps will be:

$$E = \sum_{i=1}^n (SL_i - (\alpha * \sqrt[4]{a_{\max,i} - a_{\min,i}} + \beta * f_i + \gamma * v_i))^2 \quad (15)$$

Taking $X = [x_1, x_2, \dots, x_n]$, where $x_i = [\sqrt[4]{a_{\max,i} - a_{\min,i}}, f_i, v_i]^T$, as the input vector, $D = [SL_1, SL_2, \dots, SL_n]^T$ as the expectation vector, then Equ.(15) can be expressed as:

$$E = (D - X^T W)^T (D - X^T W) \quad (16)$$

where $W = (\alpha, \beta, \gamma)^T$ are the estimated step parameters. At last, according to the least mean square algorithm, W can be calculated as follows:

$$W = [X X^T]^{-1} [X D] \quad (17)$$

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

To evaluate our proposed method, a series of experiments were conducted in the first and sixth buildings of Hangzhou Dianzi University, whose indoor layout is shown in Figure 10.

TABLE IV
USER PROFILE

USER No.	GENDER	HEIGHT (CM)	WEIGHT(KG)	PHONE
1	F	155	49	MATE 8
2	M	168	74	NEXUS 5
3	M	168	60	MATE 8
4	M	178	70	NEXUS 5
5	M	170	66	MATE 8
6	M	172	64	NEXUS 5
7	M	181	69	MATE 8
8	F	162	52	NEXUS 5

The experimental data collection method is similar to that of [29]. Eight volunteers were asked to collect data using two smartphones (Huawei Mate 8 and Google Nexus 5). Table IV shows the volunteers' specific information. Both smartphones have built-in 3-axis accelerometers and 3-axis magnetometers with a sampling frequency of 50 Hz. During the data collection, the volunteers were asked to remember how many steps they took in each walking pattern. The data collection includes the training and testing data collections. In the process of collecting the training data, each volunteer was asked to walk at a constant speed for a total of 100 steps in three walking patterns (MP, NW, and QW) using two phone poses (Pose1 and Pose2) along two paths. We did not strictly measure the volunteers' walking speeds, and we only asked them to perform MP, NW, and QW according to their own habits. Therefore, there are 96 trajectories, each of which corresponds to one pattern, one phone pose, and one path. When a volunteer walks at a constant pace, his/her step length for each step is approximately the same. The ground-truth step length for the SLE can then be obtained by dividing the length of the path by the number of steps walked. In the testing data collection, the volunteers were asked to perform a continuous motion with a fixed number of steps along another path three times. The continuous motion consists of three walking patterns using two phone poses. Therefore, there are 24 trajectories, each including three walking modes and two phone poses. We use a tuple $\langle \text{walking pattern, phone pose, the number of steps} \rangle$ for describing a part of the continuous motion. Therefore, the order of the walking patterns in the testing path is $\langle \text{NW, Pose1, 20} \rangle \rightarrow \langle \text{MP, Pose1, 20} \rangle \rightarrow \langle \text{NW, Pose1, 20} \rangle \rightarrow \langle \text{QW, Pose1, 20} \rangle \rightarrow \langle \text{MP, Pose2, 30} \rangle \rightarrow \langle \text{NW, Pose2, 30} \rangle \rightarrow \langle \text{QW, Pose2, 30} \rangle$.

B. Step Detection Evaluation

We compared the performance of our proposed SD method with Zhang's [13] and Gu's algorithms [48]. The former combines the zero-crossing and peak detection methods to limit the interval between two zero points and reduce the incidence of false zero points. The latter applied periodicity, similarity, and continuity to the peak detection to eliminate false peaks and to improve the step-counting accuracy. In addition, the parameters of our DTW-based SD method are determined according to the training data, as shown in Table V. These parameters are independent of the pedestrian, smartphone, and walking pattern but related to the sensor's sampling frequency.

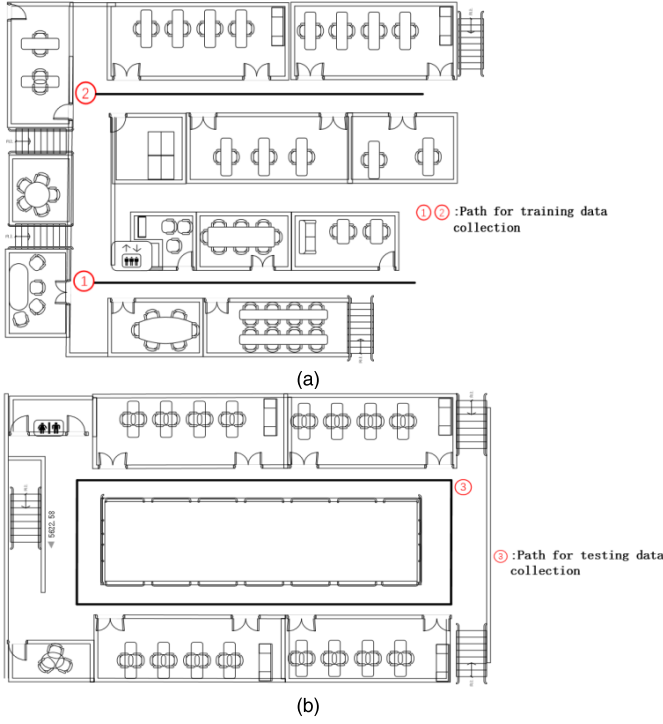


Fig. 10. The indoor layout of the experiment site. (a) The training data collection occurred in the first building. (b) The testing data collection occurred in the sixth building.

TABLE V
PARAMETER SETTING OF OUR DTW-BASED SD METHOD

Parameters	value
φ	40
τ	20
θ	0.4

We adopted the relative error to compare the SD performance, which is defined as follows:

$$SD_e = \frac{|N_e - N_t|}{N_t} \times 100\% \quad (18)$$

where N_e and N_t are the estimated value and the true value, respectively. We calculated the average relative error of all the testing trajectories, and the results are shown in Figure 11.

Figure 11 shows that our SD method is better than the two comparison methods. Zhang's method is obviously not suitable for MP because the MP error rate is significantly higher than that of the other two walking patterns. In addition, the change in smartphone pose during walking invalidated the threshold in Zhang's method, which led to the error in the QW pattern. Although Gu's method has an acceptable detection accuracy in the NW pattern, its SD results worsened in the MP and QW patterns. Therefore, both Zhang's and Gu's methods cannot adapt to different walking patterns. Our proposed method uses DTW for peak prediction, combines periodic constraints and adaptive thresholds to adapt to different walking patterns, and

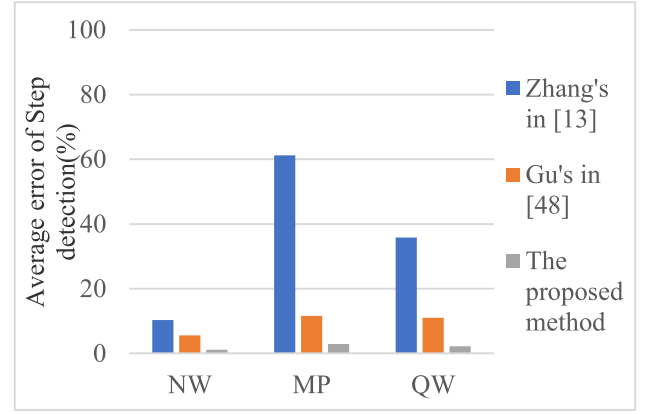


Fig. 11. Step detection performance comparison.

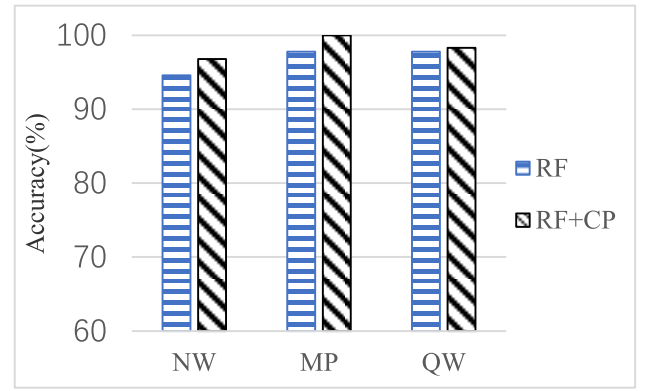


Fig. 12. Walking pattern recognition performance.

performs best in all three walking patterns, whose errors are only 1.1%, 2.9%, and 2.2%, respectively.

C. Walking Pattern Recognition Evaluation

The WPR results in Figure 12 show that RF can achieve a 96.8%, 100%, and 98.3% average recognition accuracy in NW, MP, and QW, respectively. Such recognition accuracy comes from the fact that our SD method can accurately detect each step boundary in different walking patterns, and the features extracted at each step have a good degree of discrimination. Moreover, CP also plays a big role, improving the recognition accuracy by 1.7% on average.

Tables VI, VII, and VIII show the confusion matrixes of classifiers 1, 2, and 3. We can see that the recognition of the smartphone pose is 100% correct. The worst case of misclassification occurs between NW and MP when the smartphone is in Pose2. This is because the distinction between NW and MP in Pose2 mainly depends on the features of separate three-axis accelerations. However, it is difficult to guarantee that the smartphone's three-axis direction has been unchanged due to the vibration during motion, and this results in the misclassification of NW and MP.

D. Stride Length Estimation Evaluation

In this experiment, we divided the previously collected NW and QW training data into two parts: one for training the SLE and

TABLE VI
THE CONFUSION MATRIX OF CLASSIFIER 1

The actual smartphone pose	The predicted smartphone pose	
	Pose1	Pose2
Pose1	100%	0%
Pose2	0%	100%

TABLE VII
THE CONFUSION MATRIX OF CLASSIFIER 2

The actual walking pattern	The predicted walking pattern		
	NW	MP	QW
NW	97.2%	2.8%	0%
MP	1.3%	98.7%	0%
QW	0.7%	2.5%	96.8%

TABLE VIII
THE CONFUSION MATRIX OF CLASSIFIER 3

The actual walking pattern	The predicted walking pattern		
	NW	MP	QW
NW	93.3%	6.7%	0%
MP	2.9%	97.1%	0%
QW	0.9%	0.8%	98.3%

parameters and the other for testing the SLE performance. In addition, the data of different phone poses and walking patterns is processed separately and, hence, there are four combination modes in total.

Our proposed SLE is compared with Weinberg's in [32] and Shin's in [36]. All three methods have the same step parameter training process and different step parameters in each combination mode. The cumulative distribution function (CDF) of the estimation error is used to indicate the estimation accuracy. As described in Figure 13, we can see that our proposed SLE's estimation error has a probability of 90% less than 0.04 m, which is better than Shin's and Weinberg's. The reason for the higher accuracy of our proposed SLE method is that it uses more characteristics for modeling.

E. Walking Distance Estimation Evaluation

In this experiment, we also adopted the relative error of the estimated walking distance, as shown in (19), to measure the performance.

$$WDe = \frac{|D_e - D_t|}{D_t} \quad (19)$$

where D_e and D_t are the estimated and the true walking distance, respectively. The estimated walking distance is calculated by summing all the steps for each testing trajectory:

$$D_e = \sum_{i=1}^N SL_i \quad (20)$$

where N is the number of steps, and SL_i is each step's stride length. Since the calculation of the walking distance requires

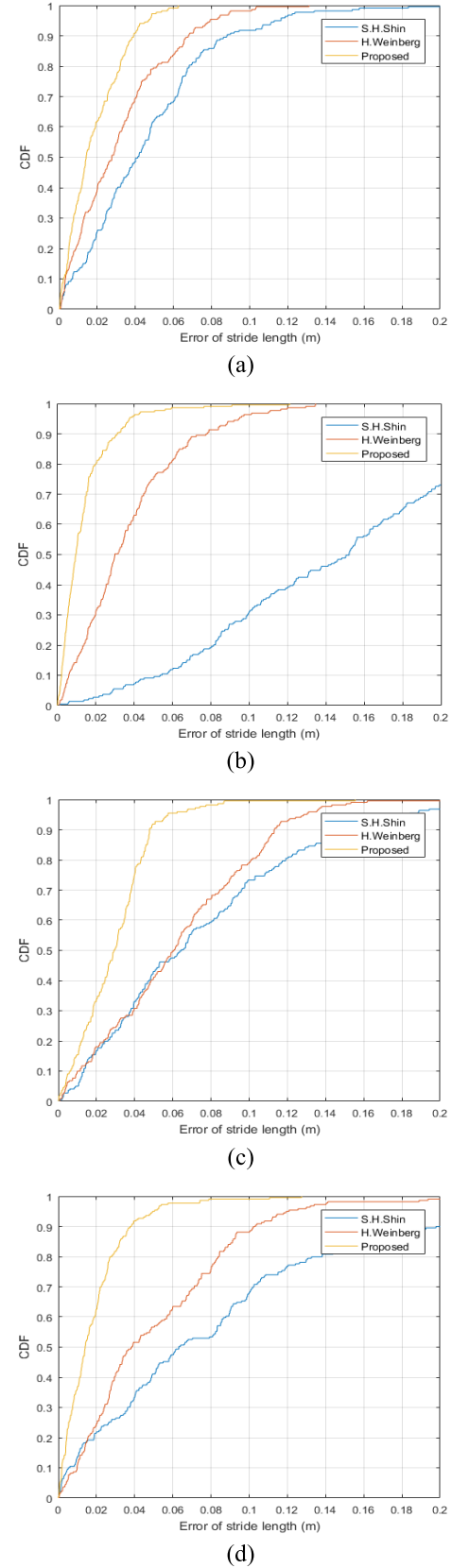


Fig. 13. The CDF of the stride length error in (a) NW in pose1, (b) QW in pose1, (c) NW in pose2, and (d) QW in pose2.

both SD and SLE, we combined Zhang's [13] and Gu's [48] SD methods with Weinberg's [32] SLE method as comparison algorithms, respectively. In addition, we also compared the

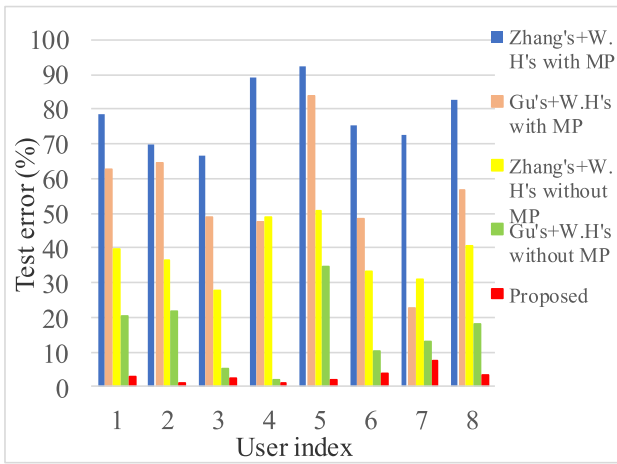


Fig. 14. Performance comparison with commonly used methods.

performance of algorithms with and without the MP mode. The final comparison results are shown in Figure 14.

From Figure 14, it can be seen that our method outperforms the commonly used walking distance estimation methods. For all the users, our method can achieve the best estimation accuracy, and its average relative error is 3.0%. Since the first two comparison algorithms cannot recognize the MP mode, their walking distance estimation errors are very large. This indicates that if the pedestrian adopts an MP pattern during the walking process, it is very important to identify MP for the walking distance estimation. Although the latter two comparison algorithms do not consider the influence of MP on the walking distance estimation, their walking distance estimation errors are still larger than that of our proposed method. The reason why our proposed method works well lies in three aspects. First, our method's SD results are more accurate, as shown in Figure 11. Second, our SLE model is a fusion of two algorithms, which is better than the comparison algorithm, as shown in Figure 13. Third, we distinguished walking patterns and trained different SLE parameters for NW and QW.

V. CONCLUSION

In this paper, we propose a robust SD and SLE method for PDR based on recognizing three walking patterns using the smartphone's built-in inertial sensors. Our proposed method includes a novel SD, an RF-based WPR, and a walking pattern-aware SLE. In particular, the proposed SD fuses DTW-based peak prediction and zero-crossing detection, and thus it can accurately identify each step's starting and ending points. Then, on the basis of the extracted features from each step, three RF-based classifications with CP are used to recognize two phone poses and three pedestrian walking patterns. According to the recognition results of the walking patterns, the proposed SLE method can adaptively select the appropriate SLE model, avoiding the large SLE errors caused by unknown walking patterns. By comparing our method with the existing commonly used methods, we show our method's superiority.

In the future, we will consider more walking patterns and phone poses, such as walking upstairs, walking downstairs, and swinging the phone in hand, to improve its application range. In addition, we will also invite more volunteers to participate in experiments to further improve our method's generalization and practicability.

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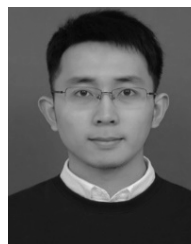
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